

# Reflection Knockoffs via Householder Reflection: Applications in Proteomics and Genetic Fine Mapping

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# Intro: Knockoff and Knockoff filter

- Constructing knockoff features  $\tilde{\mathbf{X}}$  from  $\mathbf{X}$ ;
- Fit  $y = f(\mathbf{X}, \tilde{\mathbf{X}}) + e$  to obtain  $\beta$  and  $\beta'$  for  $\mathbf{X}$  and  $\tilde{\mathbf{X}}$ .
- Compute *knockoff statistics*  $W_j = |\beta_j| - |\tilde{\beta}_j|$ .
- Knockoff threshold  $T$  for  $W$  at a nominal FDR level  $q$  is:

$$T = \min \left\{ t : \frac{\#\{j : W_j \leq -t\}}{\#\{j : W_j \geq t\} \vee 1} \leq q \right\}. \quad (1)$$

## Intro: Knockoff and Knockoff filter

For the knockoffs to have the desired property of controlling FDR,  $\mathbf{X}$  and  $\tilde{\mathbf{X}}$  must follow:

$$\text{cov}([\mathbf{X}, \tilde{\mathbf{X}}]) = \begin{bmatrix} \Sigma & \Sigma - \mathbf{S} \\ \Sigma - \mathbf{S} & \Sigma \end{bmatrix} \quad (2)$$

where  $\mathbf{S} \geq \mathbf{0}$  is a diagonal matrix that satisfies

$$2\mathbf{S} - \mathbf{S}\Sigma^{-1}\mathbf{S} \geq \mathbf{0}. \quad (3)$$

$\mathbf{S}$  controls the degree of uncorrelated-ness between  $\mathbf{X}_j$  and  $\tilde{\mathbf{X}}_j$ , and larger  $\mathbf{S}$  values bring more power to the knockoff filter.

# Intro: Estimating $S$

- Current method use semidefinite program (SDP) to estimate  $S$ .
- We introduce a novel method to estimate  $S$  via Householder reflection.

# Reflection

## Lemma

Suppose the data matrix  $\mathbf{X}$  is  $n \times p$  with  $n > p$  and  $\mathbf{X}$  is column-centered, scaled, and has full rank  $p$ . Denote  $\mathbf{P}_j = \mathbf{X}_{,-j}(\mathbf{X}_{,-j}^t \mathbf{X}_{,-j})^{-1} \mathbf{X}_{,-j}^t$  where  $\mathbf{X}_{,-j}$  is matrix  $\mathbf{X}$  with  $j$ -th column removed. Compute  $\mathbf{Y}$  such that its  $j$ -th column is a Householder reflection of  $\mathbf{X}_j$  with respect to  $\mathbf{P}_j$ ,  $\mathbf{Y}_j = (2\mathbf{P}_j - \mathbf{I})\mathbf{X}_j$ . Define  $\mathbf{Z} = \frac{1}{2}(\mathbf{X} - \mathbf{Y})$ . Denote  $\text{cov}(\mathbf{X}_j, \mathbf{Y}_k)$  as sample covariance between vector  $\mathbf{X}_j$  and  $\mathbf{Y}_k$ . Then the following hold:

- a)  $\text{cov}(\mathbf{Y}_j, \mathbf{X}_k) = \text{cov}(\mathbf{X}_j, \mathbf{X}_k)$  for all  $j \neq k$  and  $j, k \in (1, 2, \dots, p)$ .
- b)  $\text{cov}(\mathbf{Y}_j, \mathbf{X}_j) < \text{var}(\mathbf{X}_j)$  for all  $j \in (1, 2, \dots, p)$ .
- c) Let  $\mathbf{T}$  be a matrix such that  $\mathbf{T}_{jk} = \text{cov}(\mathbf{X}_j, \mathbf{Z}_k)$ , then  $\mathbf{T}_{jk} = 0$  for all  $j \neq k$  and  $\mathbf{T}_{jj} = \text{cov}(\mathbf{Z}_j, \mathbf{Z}_j) > 0$  for all  $j$ .
- d)  $\mathbf{X}(\mathbf{X}^t \mathbf{X})^{-1} \mathbf{X} \mathbf{Y} = \mathbf{Y}$  and  $\mathbf{X}(\mathbf{X}^t \mathbf{X})^{-1} \mathbf{X} \mathbf{Z} = \mathbf{Z}$ .

# Construction

the mean  $\mathbf{b}$  and covariance  $\mathbf{C}^t\mathbf{C}$  of a knockoff  $\tilde{\mathbf{X}}$  conditional on  $\mathbf{X}$  are given by

$$\begin{aligned}\mathbf{b} &= \mathbf{X}(\mathbf{I} - \Sigma^{-1}\mathbf{S}) \\ \mathbf{C}^t\mathbf{C} &= 2\mathbf{S} - \mathbf{S}\Sigma^{-1}\mathbf{S}.\end{aligned}\tag{4}$$

Let  $\mathbf{S} = \alpha\mathbf{T}$  for  $\alpha \in (0, 1]$ . Straightforward calculations using the identities in Lemma 1 parts (c) and (d) yields:

$$\begin{aligned}\mathbf{b} &= \mathbf{X} - \alpha\mathbf{Z} \\ \mathbf{C}^t\mathbf{C} &= 2\alpha\mathbf{T} - \alpha^2 \text{cov}(\mathbf{Z}).\end{aligned}\tag{5}$$

## Construction

Let  $\mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^t$  be the eigendecomposition of  $\text{cov}(\mathbf{Z}\mathbf{T}^{-1/2})$ . Substituting this, we obtain

$$\mathbf{C}^t\mathbf{C} = \mathbf{T}^{1/2}\mathbf{Q} [2\alpha\mathbf{I} - \alpha^2\mathbf{\Lambda}] \mathbf{Q}^t\mathbf{T}^{1/2}. \quad (6)$$

For  $\mathbf{C}^t\mathbf{C}$  to be positive semi-definite, it is sufficient to have

$$\mathbf{B}^2(\alpha) \triangleq 2\alpha\mathbf{I} - \alpha^2\mathbf{\Lambda} > \mathbf{0}. \quad (7)$$

Therefore it's sufficient to have  $\alpha \leq 2/\Lambda_1$ , where  $\Lambda_1$  is the leading eigenvalue of  $\text{cov}(\mathbf{Z}\mathbf{T}^{-1/2})$ . Since  $\Lambda_1$  is positive and finite, and  $\alpha \in (0, 1]$ , the optimal  $\hat{\alpha}$  exists. This ensures  $\mathbf{C}^t\mathbf{C}$  is positive semi-definite, with  $\mathbf{C} = \mathbf{B}(\hat{\alpha})\mathbf{Q}^t\mathbf{T}^{1/2}$ . Consequently,  $\mathbf{S} = \hat{\alpha}\mathbf{T} \leq 2 \text{cov}(\mathbf{X})$ , as guaranteed by the Schur complement condition.

We thus construct the knockoff as

$$\tilde{\mathbf{X}} = \mathbf{X} - \hat{\alpha}\mathbf{Z} + \mathbf{U}\mathbf{B}(\hat{\alpha})\mathbf{Q}^t\mathbf{T}^{1/2} \quad (8)$$

where: If  $\mathbf{U}$  is chosen from the null space of  $\mathbf{X}$ ,  $\tilde{\mathbf{X}}$  is akin to Fixed-X knockoffs. If  $\mathbf{U}_j \sim \text{MVN}(\mathbf{0}, \mathbf{I}_n)$ ,  $\tilde{\mathbf{X}}$  is akin to Model-X second-order knockoff.

## Shrinkage priors and data augmentation

In Lemma 1, computing  $\mathbf{Y}_j$  requires the least square fit of the linear regression:

$$\mathbf{X}_j = \mathbf{X}_{,-j}\beta_{-j} + \mathbf{e} \quad (9)$$

to obtain  $\hat{\beta}_{-j}$ , which is then used to calculate  $\mathbf{Y}_j = 2\mathbf{X}_{,-j}\hat{\beta}_{-j} - \mathbf{X}_j$ . When  $p$  is large compare to  $n$ , and/or there exist highly correlated features, the model (9) is prone to overfitting, which can adversely affect the performance of the knockoff filter.

Suppose we put priors on  $\beta_{-j}$  such as

$$\beta_{-j} \sim MVN(0, \sigma^2 \mathbf{I}_{(p-1)}), \quad (10)$$

where  $MVN$  denotes multivariate normal distribution,  $\mathbf{I}_m$  is the identity matrix of dimension  $m$ , and  $\sigma$  is the prior effect size to be specified. The Bayesian estimated effect sizes are given by

$$\hat{\beta}_{-j} = (\mathbf{X}_{,-j}^t \mathbf{X}_{,-j} + \sigma^{-2} \mathbf{I}_{(p-1)})^{-1} \mathbf{X}_{,-j}^t \mathbf{X}_j. \quad (11)$$

The same solution can be achieved via augmenting  $\mathbf{X}$  with ghost observations  $\sigma^{-1} \mathbf{I}_p$  such that  $\mathbf{X} \leftarrow [\mathbf{X}^t, \sigma^{-1} \mathbf{I}]^t$ .



## EB estimates of shrinkage priors

$$\begin{aligned}\mathbf{X}_j &= \mathbf{X}_{,-j}\beta + \mathbf{e} \\ \beta &\sim MNV(0, \tau^{-1}\sigma^2) \\ \mathbf{e} &\sim MNV(0, \tau^{-1}\mathbf{I}_n)\end{aligned}\tag{12}$$

Integrate out  $\beta$  to obtain the marginal likelihood, and we get

$$\mathbf{X}_j | \mathbf{X}_{,-j}, \sigma^2 \sim MNV(0, \tau^{-1}\sigma_j^2 \mathbf{X}_{,-j} \mathbf{X}_{,-j}^t + \tau^{-1}\mathbf{I}).\tag{13}$$

This is a special case of a linear mixed model and we can obtain maximum likelihood estimate of  $\sigma_j^2$  using IDUL [?]. To proceed, we obtain eigendecomposition  $\mathbf{X}_{,-j} \mathbf{X}_{,-j}^t = \mathbf{Q}_j \mathbf{D}_j \mathbf{Q}_j^t$  where  $\mathbf{Q}_j$  is orthonormal and  $\mathbf{D}_j$  is diagonal. We rotate the system by multiplying (12) by  $\mathbf{Q}^t$  from left we get

$$\mathbf{X}'_j | \mathbf{X}'_{,-j}, \sigma^2 \sim MNV(0, \tau^{-1}\sigma_j^2 \mathbf{D}_j + \tau^{-1}).\tag{14}$$

# Computation: QR decomposition for Householder reflection

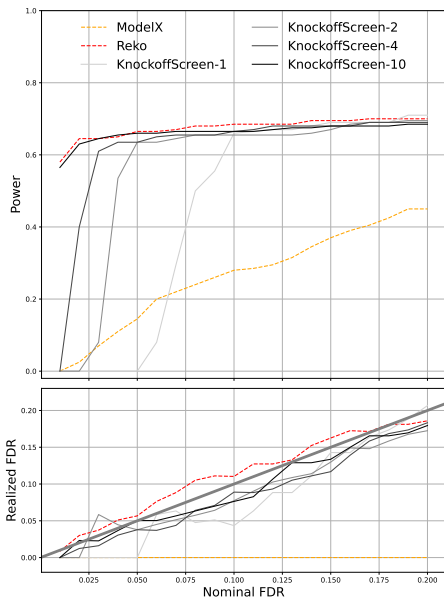
- 1: **Input:** A matrix  $\mathbf{X}$  of  $n \times p$  with full rank  $p$ .
- 2: **Output:**  $\mathbf{Y}$  of  $n \times p$ .
- 3: Augment  $\mathbf{X}$  by appending  $\Sigma_p$ , update  $n \leftarrow (n + p)$ .
- 4: Perform QR decomposition to obtain  $\mathbf{X} = \mathbf{QR}$ .
- 5: **for** each column  $\mathbf{R}_j$  of  $\mathbf{R}$  **do**
- 6:     Denote  $\mathbf{R}_{,-j}$  as matrix  $\mathbf{R}$  with  $j$ -th column removed.
- 7:     Do Givens rotation on  $\mathbf{R}_{,-j}$  to obtain an upper triangular matrix  $\mathbf{U}$ .
- 8:     Do the same sequence of Givens rotation on  $\mathbf{R}_j$  to obtain feature  $\mathbf{V}$ .
- 9:     Solve  $\mathbf{V} = \mathbf{U}\beta_{-j}$  with backward substitution to get  $\hat{\beta}_{-j}$ .
- 10:    Compute  $\tilde{\mathbf{R}}_j = 2\mathbf{R}_{,-j}\hat{\beta}_{-j} - \mathbf{R}_j$ , which is  $j$ -th column of  $\tilde{\mathbf{R}}$ .
- 11: **end for**
- 12: Compute  $\mathbf{Y} = \mathbf{Q}\tilde{\mathbf{R}}$
- 13: **return**  $\mathbf{Y}[1 : n, ]$

# Aggregation of multiple knockoff statistics

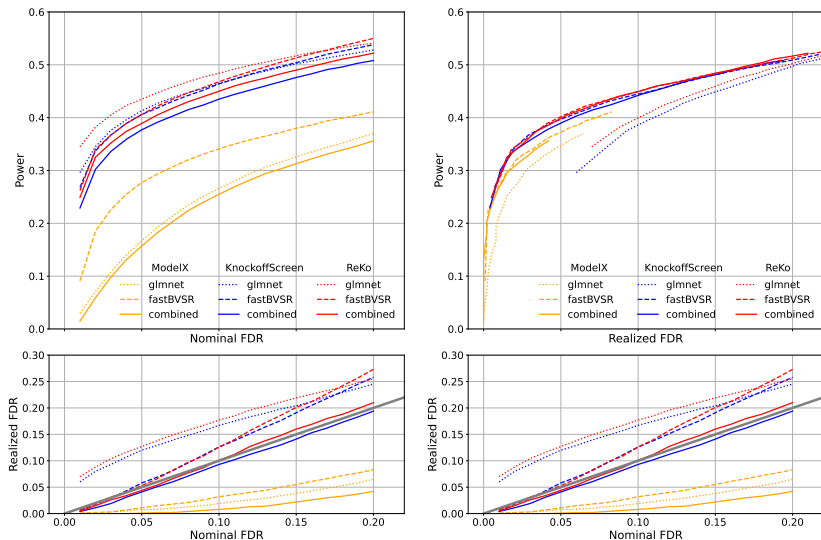
## Lemma

*Let  $W_{\text{cat}} = (W^{(1)}, \dots, W^{(M)})$ , i.e., the pooled knockoff statistics from the  $M$  runs. If we apply the standard knockoff threshold selection rule (1) to  $W_{\text{cat}}$ , the resulting threshold  $T_{\text{cat}}$  also controls the FDR at level  $q$ . Define the average knockoff statistic  $\bar{W}_j = \frac{1}{M} \sum_{m=1}^M W_j^{(m)}$ . Selecting feature  $j$  such that  $\bar{W}_j > T_{\text{cat}}$  controls the FDR at level  $q$ .*

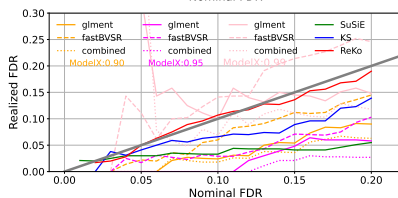
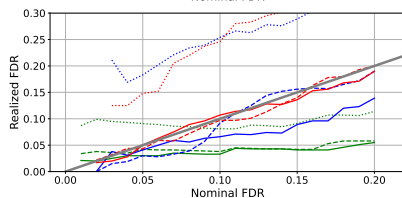
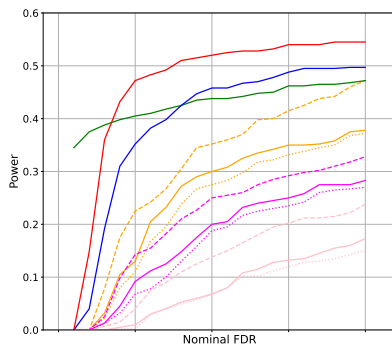
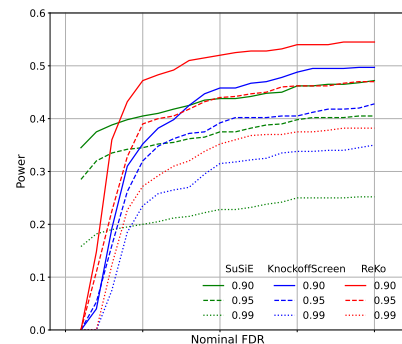
# Aggregation



# Comparison: Proteomics from UKBB



# Comparison: HLA-DQB1 from FHS



## Proteomic association with age

| Level | MX  | KS  | ReKo | Overlap   |
|-------|-----|-----|------|-----------|
| 0.05  | 256 | 306 | 309  | 291 (90%) |
| 0.10  | 308 | 361 | 371  | 345 (89%) |

# Data

| eGene         | eQTL                      | MX    | KS    | ReKo | SuSiE | Type |
|---------------|---------------------------|-------|-------|------|-------|------|
| <i>PPIEL</i>  | rs79473113:1:39554348:1   | 1.00  | 1.00  | 1.00 | 1.00  | 1111 |
|               | rs7520271:1:39579041:1    | 0.80  | 0.82  | 0.87 | 0.97  | 1111 |
| <i>GJB6</i>   | rs12875121:13:20478719:1  | 0.95  | 0.97  | 0.98 | 0.99  | 1111 |
|               | rs12875121:13:20478719:0  | 0.76  | 0.78  | 0.79 | 0.89  | 1110 |
|               | rs9506450:13:20239478:0   | -0.15 | -0.27 | 0.61 | 0.44  | 0010 |
|               | rs9579911:13:20637437:0   | 0.23  | 0.26  | 0.38 | 0.20  | 0010 |
| <i>CCR9</i>   | rs9814578:3:45821775:1    | 0.85  | 0.93  | 0.98 | 0.98  | 1111 |
| <i>COA8</i>   | rs10143623:14:103720539:0 | 0.66  | 0.58  | 0.85 | 0.87  | 1110 |
| <i>NECAB3</i> | rs2626556:20:33768509:0   | 0.79  | 0.80  | 0.88 | 0.95  | 1111 |



# Accomplishments

- Developed ReKo.
- Aggregating multiple knockoff statistics.
- The benefits of  $\otimes$ fastBVSR and  $\otimes$ combined.

# Challenges

- Extremely high correlation. Perhaps clustering first?
- Reflection over subspace spanned by leading eigenvectors?
- ReKo for data sharing.